

GenSec Industries: Protocol specification document



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1. Introduction

The focus of this document is to keep track of all the protocol decisions that were made for the occupancy management system. Information regarding the communication protocol, message tables and any additional information regarding the protocol aspect of the system can be found here.

2. Protocol:

The MQTT protocol will be used for this project. The system is structured by using multiple different topics. Each with their own responsibilities such as keeping track of the node being alive or for basic communication. See section 3.1 for additional details.

3. Messages rules:

This section of the document will give more information on the communication protocol and why it was chosen. See Analys document more information.

3.1: Communication basis:

By leveraging a network of sensors and devices, the system allows users to remotely control and monitor various aspects of the security network. It employs a robust communication protocol to ensure seamless data exchange between devices and easy to understand commands that will help with debugging. The communication's core objective is to create an integrated, efficient, and secure solution. See table 1 for the symbol that are described in this section.

Table 1: Protocol Symbols

#	Symbol Name	Symbol
1	Start Symbol	"&"
2	End Symbol	","
3	Send-to Symbol	">"
3	Separation Symbol	"_"
4	Add Symbol	"+"
5	Multiple instances	" , "

Commands are tied to an ID, and later parsed as an **ENUM**.

To know which device is communicating in the system, there will have to instances of ID's. Since it is not expected to have more than 1000 devices connected at the same time, the system will operate on a **3-digit ID** (e.g. 045). When a message is being broadcasted and no destination ID is required, the 3 digits will be 000. If the destination is **required**, the ">" symbol is used.

To separate the ID and the rest of the message, a **separation character** must be used. It can also be used in a different instance of a message.

Data sometimes needs to be added to a certain aspect of the system. For example, when 2 people need to be added to a gate queue after scanning their tickets. To do this, an **add symbol** is needed. If multiple instances of data need to be added, a comma " , " will be used.

Here are some examples and the meaning behind them:

Table 2: Message examples.

#	Message frame	Description
1	&004>122-1;	The divider with ID 004, tells the gate with ID 122 to open.
2	&112-000-3+15;	The gate with ID 112 announces to everyone that 15 people are waiting on the line.
3	&234-021-5+2;	The divider with ID 234 allocates to people to the gate with ID 021.
4	&001-000-6;	The gate with ID 1 is searching for a divider to register to
5	&351-000-7+2;	The scanner with ID 351 reports that 2 people are waiting for a gate.

3.2: Message table:

In section 3.1 there is a mention of “commands” that all devices use. Each command has a value that represents itself. The following table shows the commands that can be used in the system:

Table 3: Commands available.

#	Command Name	Command description	Command ID
1	OPENGATE	Requests a gate to be opened.	“1”
2	CLOSEGATE	Requests a gate to be closed	“2”
3	NUMOFPEOPLE	Announces amount of people	“3”
4	EMERGENCY	Requests all devices to go into emergency mode	“4”
5	ALLOC	Dividers send a certain amount of people to a gate	“5”
6	REGISTER	Gate registers itself as being active	“6”
7	SCANNED	Scanner reports how many people are scanned and waiting for a response.	“7”